

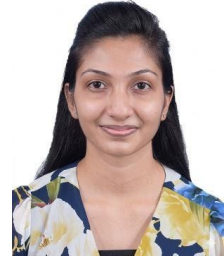
W. A. SANUJI DEWMINI WEERASINGHE

Data Science Undergraduate | University of Peradeniya

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portfolio-sanuji-s-projects.vercel.app



PROFESSIONAL PROFILE

Final-year Data Science undergraduate at the University of Peradeniya with a strong foundation in statistics, machine learning, and data engineering. Experienced in building end-to-end data solutions and deep learning models, with hands-on research in time-series forecasting. Seeking a Data Science / Data Engineering internship to leverage analytical skills and contribute to intelligent automation or business intelligence solutions.

EDUCATION

BSc (Hons) in Data Science — University of Peradeniya

2022 – Present

Current GPA: 3.45 / 4.00

G.C.E. Advanced Level

2020

Combined Mathematics: A | Chemistry: B | Physics: C

WORK EXPERIENCE

Bank of Ceylon — Job Skills Development Trainee

Jan – Apr 2021

Ratmalana Super Grade Branch | School Leavers Programme | Front & Back Office Operations

- Gained hands-on exposure to commercial banking operations by supporting customer-facing and back-office workflows within a high-volume branch environment.
- Supported daily banking operations and administrative processes, contributing to efficient branch service delivery.

PROJECTS

Monsoon-Stratified Rooftop Solar PV Forecasting for Grid Stability

2026 – Present

Final-Year Research Project (DSC 4996) | Python | PyTorch | CNN-LSTM & Transformer | ResUNet / YOLOv8-seg | GIS | Time-Series Forecasting

- Forecasted district-level rooftop solar generation across multiple monsoon conditions to support grid-stability planning.
- Estimated PV potential across 5 Sri Lankan districts via an automated satellite-imagery pipeline using YOLOv8-seg and ResUNet.
- Created a leakage-safe 39-feature dataset spanning 8 years from NASA POWER with strict chronological validation.

AI-Based Medical Claims Validation System (Janashakthi Insurance)

2025

Python | Django REST Framework | React (TypeScript) | OpenCV

- Developed a multi-portal insurance claims platform automating document verification and fraud screening across Branch, Digital, and Admin portals.
- Reduced manual review effort through a weighted five-factor AI risk-scoring engine that auto-approved, flagged, or rejected claims.
- Implemented secure role-based access control using JWT authentication across all portals.

Business Intelligence Framework for Sales Performance

2024

Power BI | SQL | ETL | Data Warehousing | OLAP

- Unified fragmented retail data from multiple branches into a centralised analytics platform, enabling organisation-wide sales visibility.
- Delivered executive Power BI dashboards and OLAP reporting tools transforming raw transactions into actionable sales insights.

Customer Churn Prediction with MLOps

2026 – Present

Python | XGBoost | MLflow | FastAPI | Docker | GitHub Actions

- Built a customer-retention analytics system identifying customers at risk of churn with SHAP-based explainability.
- Operationalized through automated CI/CD pipelines, MLflow tracking, FastAPI serving, Docker, and data-drift monitoring.

Retrieval-Augmented Q&A System (RAG)

2026

Python | ChromaDB | Sentence-Transformers | Streamlit

- Enabled natural-language querying of large document collections via semantic search, vector embeddings, and citation-backed responses.
- Deployed an interactive Streamlit interface with retrieval-augmented generation for improved answer traceability.

Tea Leaf Disease Classification

2025

Python | PyTorch | VGG16 / ResNet50 | scikit-learn

- Automated tea leaf disease identification using VGG16 and ResNet50 transfer learning, reducing dependence on manual visual inspection.
- Selected best-performing architecture through precision, recall, F1-score, and confusion-matrix analysis.

PHM Tweet Classification

2025

Python | TensorFlow / Keras | LSTM | Bi-LSTM | scikit-learn

- Developed social-media classifiers detecting public-health content from highly imbalanced datasets using LSTM and Bi-LSTM architectures.
- Improved robustness via class weighting, threshold optimisation, and cross-validation.

TECHNICAL SKILLS

Machine Learning: PyTorch | TensorFlow / Keras | XGBoost | scikit-learn | CNN | LSTM / Bi-LSTM | Transformers | Transfer Learning

Data Engineering & MLOps: PostgreSQL | MySQL | ETL | Data Warehousing | Docker | MLflow | FastAPI | GitHub Actions

Analytics & BI: SQL | Power BI | OLAP | Streamlit | Matplotlib | Seaborn

NLP & Generative AI: RAG | Embeddings | ChromaDB | Sentence-Transformers

Computer Vision & Geospatial: YOLOv8-seg | ResUNet | U-Net | OpenCV | Satellite Imagery | GIS | OCR

Languages & Tools: Python | SQL | R | Java | Git | GitHub | Jupyter

SOFT SKILLS

Leadership | Teamwork | Effective Communication | Critical Thinking | Problem Solving | Analytical Reasoning | Time Management

ACHIEVEMENTS & AWARDS

- Champion — IEEE INSL Central Provincial New Innovation Competition (Hope Project), 2024.
- Champion — KickStart Weekend 2024, SIREN Business Idea Competition (Business Idea Pitch), 2024.
- Research & Development Director — DataEX Society, University of Peradeniya, 2026.

REFERENCES

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